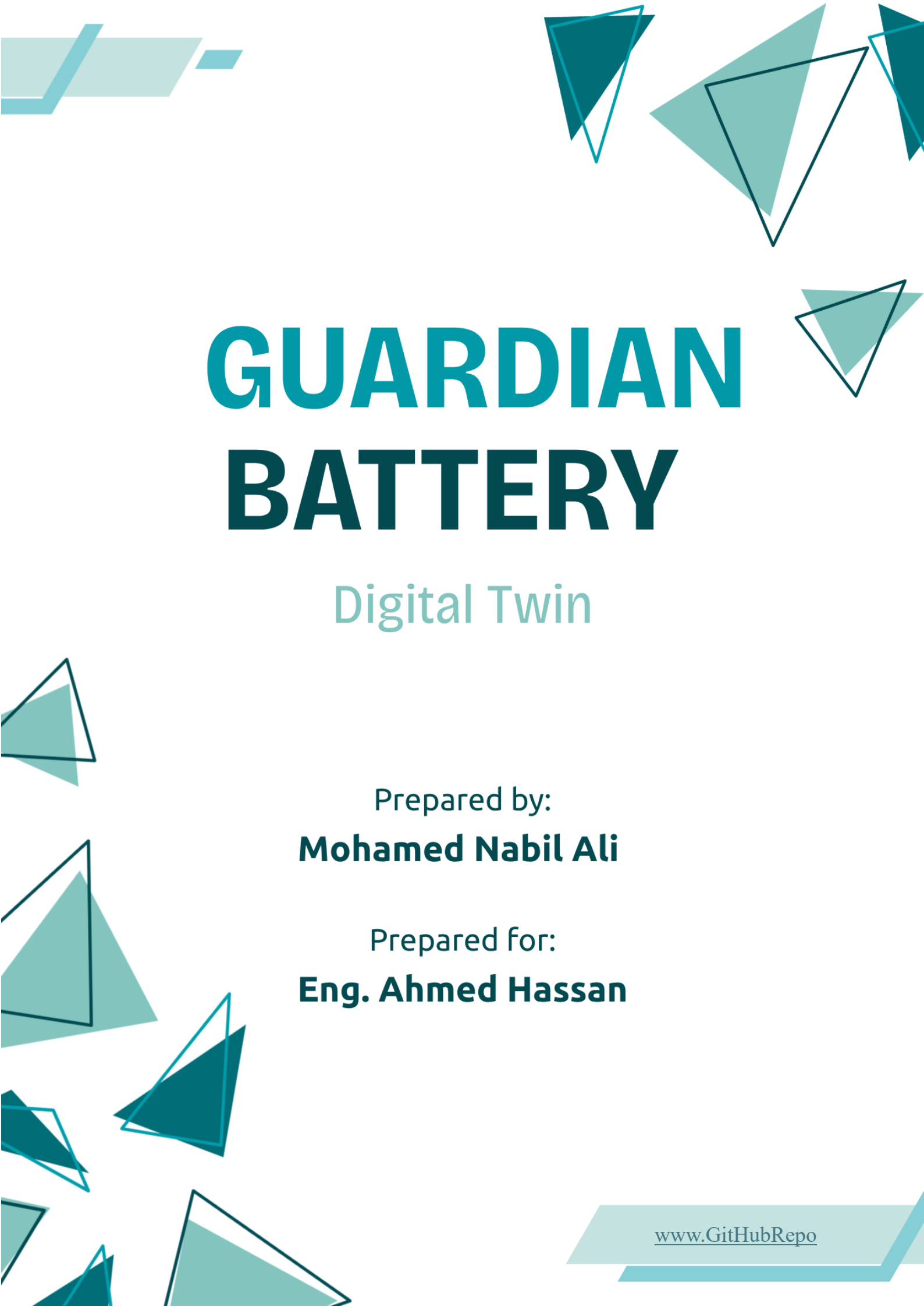


The background of the slide is decorated with various teal-colored geometric shapes, including triangles and polygons of different sizes and orientations, some with solid fills and others as outlines.

GUARDIAN BATTERY

Digital Twin

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Introduction

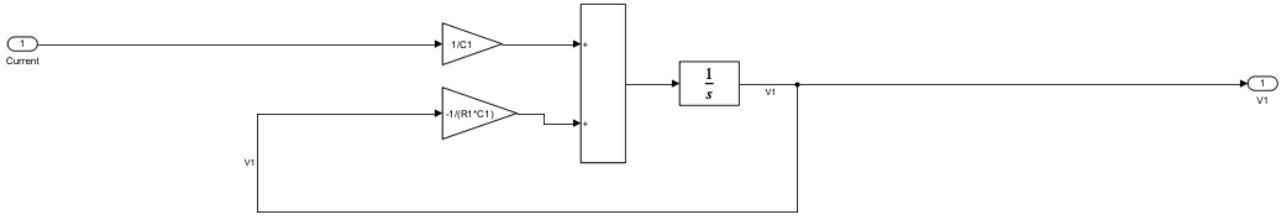
Energy storage systems have evolved rapidly, with Lithium Iron Phosphate (LiFePO_4) batteries emerging as the preferred choice for electric vehicle (EV) and grid storage applications due to their high thermal stability, safety, and extended lifespan. To ensure the efficient and safe operation of these batteries, a Battery Management System (BMS) relies on precise mathematical models that simulate the internal dynamic behavior of the cell. The Equivalent Circuit Model (ECM) is widely considered one of the most effective approaches, as it strikes an optimal balance between computational accuracy and mathematical complexity. This report focuses on the development, simulation, and analysis of an ECM for a 50Ah EVE LF50K cell, integrating advanced closed-loop algorithms to accurately estimate the State of Charge (SOC) and State of Health (SOH) under dynamic operating conditions.

Objectives

The primary objective of this project is to design and simulate a comprehensive dynamic model of a lithium-ion battery within the MATLAB/Simulink environment. This encompasses the following specific sub-objectives:

- **Core Model Development:** Construct a continuous-time first-order Equivalent Circuit Model (1RC Thevenin) to simulate the terminal voltage of the battery, incorporating hysteresis noise to represent realistic, unmodeled environmental conditions.
- **Baseline SOC Estimation:** Implement Coulomb Counting as an open-loop method to track the remaining battery capacity based on current integration.
- **Dynamic Response Enhancement (Bonus 1):** Upgrade the core architecture to a second-order Equivalent Circuit Model (2RC) to capture the battery's fast transient responses with higher fidelity.
- **Advanced State Monitoring (Bonus 2):** Integrate an Extended Kalman Filter (EKF) as a closed-loop observer. This algorithm dynamically corrects the accumulated drift inherent in open-loop Coulomb counting by minimizing the voltage residual, providing a highly robust SOC estimation.
- **State of Health (SOH) Evaluation (Bonus 3):** Apply the Total Least Squares (TLS) algorithm via Singular Value Decomposition (SVD) to estimate the true internal resistance (R_0) independent of sensor noise. This extracted parameter is then compared against the manufacturer's datasheet to evaluate the battery's overall State of Health (SOH).

Continuous-Time Plant Modeling: Polarization Voltage (V_1)



The dynamic transient behavior of the LiFePO₄ cell is characterized by its internal polarization, which is modeled using an RC network. The voltage across this branch, denoted as V_1 , dictates how the battery's voltage exponentially approaches its steady-state value when subjected to a load step.

Mathematically, the continuous-time dynamics of this RC branch are governed by the following first-order differential equation:

$$\frac{dV_1}{dt} = \frac{I}{C_1} - \frac{V_1}{R_1 \cdot C_1}$$

Where:

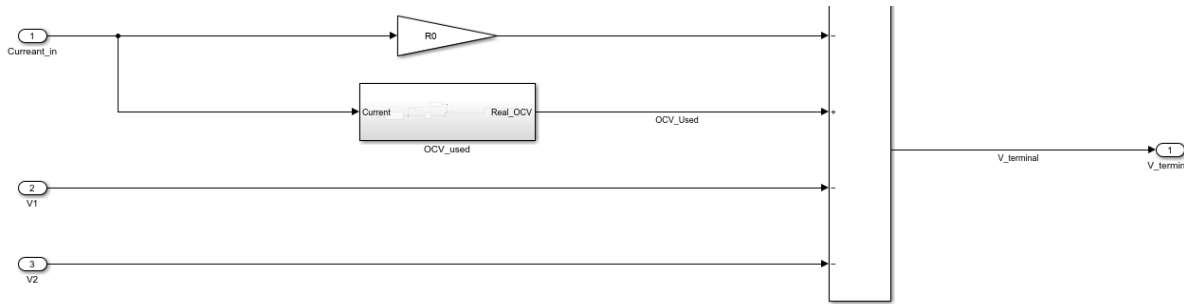
- I is the applied current (with discharge defined as positive).
- C_1 is the polarization capacitance.
- R_1 is the polarization resistance.

Simulink Implementation: To translate this differential equation into the MATLAB/Simulink environment, a state-variable approach was utilized. As shown in the provided block diagram, the implementation relies on a continuous-time integrator loop:

1. The input Current signal is multiplied by a gain block of $1/C_1$.
2. The output voltage V_1 is fed back into the system through a negative feedback loop with a gain of $-1/(R_1 \cdot C_1)$.
3. These two scaled signals are combined in a sum block and passed through a continuous-time Integrator block ($1/s$) to continuously solve for the instantaneous polarization voltage $V_1(t)$.

This modular subsystem approach not only ensures accurate mathematical integration but also allows for easy expansion, such as duplicating the architecture to create the V_2 branch for the 2RC model.

Terminal Voltage Equation and OCV Modeling



The core of the Equivalent Circuit Model is the calculation of the battery's terminal voltage. This parameter represents the actual measurable voltage at the battery's terminals under load conditions and is derived by subtracting the internal voltage drops from the theoretical Open Circuit Voltage (OCV).

For the enhanced 2RC model, the governing equation for the terminal voltage is defined as:

$$V_{terminal} = OCV(SOC) - I \cdot R_0 - V_1 - V_2$$

Where:

- $OCV(SOC)$ is the Open Circuit Voltage, which is a non-linear function of the State of Charge.
- $I \cdot R_0$ represents the instantaneous ohmic voltage drop across the series internal resistance.
- V_1 and V_2 represent the transient polarization voltage drops across the two RC branches.

Simulink Implementation: The translation of this algebraic equation into the model is visually demonstrated in the terminal voltage subsystem. The implementation follows a straightforward routing logic:

1. **Ohmic Drop:** The input current signal is routed through a Gain block configured with the parameter R_0 to calculate the instantaneous $I \cdot R_0$ voltage drop.
2. **OCV Generation:** A dedicated subsystem (OCV_used) evaluates the current State of Charge and outputs the corresponding true baseline voltage (Real_OCV) using a 1D Lookup Table.
3. **Signal Summation:** The final mathematical operation is executed using a multi-input Sum block. This block is configured with the specific sign array $+ - -$, where the OCV acts as the positive base value, and the ohmic drop ($I \cdot R_0$), along with the polarization voltages (V_1 and V_2), are subtracted from it to yield the final $V_{terminal}$ output.

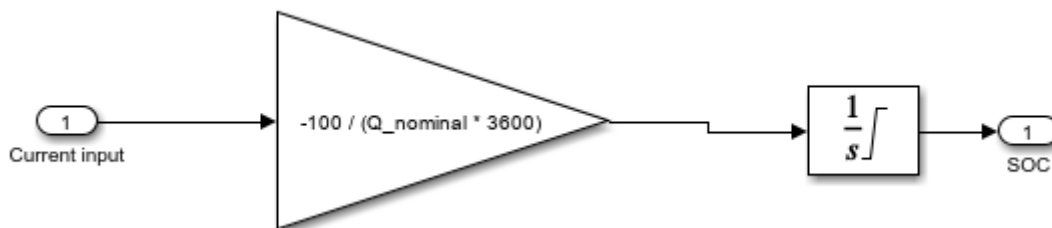
State of Charge (SOC) Tracking and OCV Mapping

To accurately determine the baseline Open Circuit Voltage (OCV), the model must first calculate the real-time State of Charge (SOC). This process, along with the injection of environmental uncertainties, is encapsulated within the OCV_used subsystem shown in the provided block diagram,

The architecture of this subsystem consists of three primary stages:

1. Coulomb Counting (Open-Loop SOC Estimation)

The first stage relies on the Coulomb Counting algorithm to track the capacity of the battery. As illustrated in the subsystem diagram, the system takes the raw Current input signal and processes it through a Gain block configured with the scaling factor $-100 / (Q_{\text{nominal}} * 3600)$. This specific factor mathematically converts the raw current (Amperes) into a percentage rate of change based on the battery's nominal capacity. The scaled signal is then fed into a continuous-time Integrator block ($1/s$), which integrates the current over time to output the instantaneous SOC. To facilitate post-processing comparisons (specifically against the EKF output), this SOC value is routed to a workspace logging block designated as *out.SOC_CC_log*.

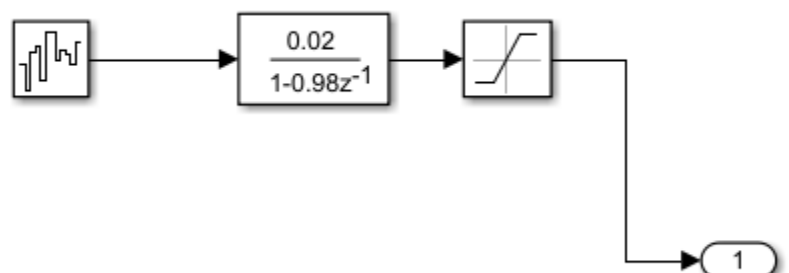


2. Non-linear OCV Lookup

characterized by a flat voltage plateau across most of its usable capacity simple linear equation cannot represent the OCV. Instead, the calculated SOC signal is fed into a 1-D Lookup Table (1-D T(u)). This block maps the dynamic SOC percentage to its corresponding theoretical Open Circuit Voltage (OCV) based on empirical breakpoint data.

3. Hysteresis Noise Injection

In a real-world scenario, a battery's voltage exhibits hysteresis (path-dependence based on whether it is charging or discharging) and is subject to unmodeled environmental noise. To simulate these non-ideal conditions and test the robustness of the system, a Hysteresis Noise subsystem generates a disturbance



signal. This noise is mathematically added to the pure OCV output from the lookup table using a sum block.

Extended Kalman Filter (EKF) Implementation



To overcome the inherent drift and error accumulation of the open-loop Coulomb Counting method, an Extended Kalman Filter (EKF) was implemented as a closed-loop state observer. Due to the highly non-linear relationship between the battery's OCV and SOC, a standard Kalman Filter is insufficient; the EKF linearizes this relationship at each time step using the local derivative of the OCV curve.

Simulink Integration:

The EKF algorithm was integrated into the model using a MATLAB Function block. The filter requires three primary inputs:

1. I_{meas} : The instantaneous load current.
2. V_{meas} : The actual terminal voltage of the battery (which includes the unmodeled hysteresis noise).
3. T_s : The discrete sample time of the system (1 second).

Correction Mechanism:

During operation, the EKF performs a prediction step based on current integration (similar to Coulomb Counting) to estimate an initial SOC and expected terminal voltage. It then compares this expected voltage against the actual measured voltage (V_{meas}). The resulting voltage residual is multiplied by the Kalman Gain to continuously correct the SOC estimate. This closed-loop feedback allows the EKF to successfully detect unmodeled voltage drops (such as the injected hysteresis noise) and dynamically adjust the SOC trajectory to reflect the true state of the battery.

Total Least Squares (TLS) and State of Health (SOH) Evaluation

To quantify the degradation of the cell, the internal series resistance (R_0) was identified online using Total Least Squares (TLS) via Singular Value Decomposition (SVD).

Unlike Ordinary Least Squares (OLS), which assumes noise exists only in the dependent variable, TLS accounts for errors in both the measured voltage and the current profile.

Mathematical Formulation:

1. The polarization voltage (V_1) was reconstructed using a discrete-time first-order expansion:

$$V_{1,k+1} = a_1 V_1(k) + R_1(1 - a_1)I_k$$

where $a_1 = e^{-T_s/(R_1 C_1)}$.

2. The regression equation was structured by isolating the ohmic drop, defining the regressor matrix D containing the centered current x_k and the derived voltage drop $y_k = OCV(SOC) - V_1 - V_{meas}$.
3. Applying SVD to D :

$$D = USV^T$$

The elements of the right-singular matrix V were used to extract the optimal resistance estimate:

$$R_{0,TLS} = -\frac{V_{1,2}}{V_{2,2}}$$

4. **State of Health (SOH):** The extracted resistance was benchmarked against the manufacturer's initial datasheet value ($R_{0,datasheet} = 0.70 \text{ m}\Omega$) to compute the percentage health ratio:

$$SOH = \left(\frac{R_{0,datasheet}}{R_{0,TLS}} \right) \times 100\%$$

```

% TLS and SOH Calculation
I_k = current_A;
V_meas_k = interp1(time_vol, voltage, time_s, 'linear', 'extrap');
SOC_CC_k = interp1(time_soc, soc_cc, time_s, 'linear', 'extrap');

Ts = 1;
a1 = exp(-Ts / (R1 * C1));
V1_hat = zeros(length(I_k), 1);
for k = 1:length(I_k)-1
    V1_hat(k+1) = a1 * V1_hat(k) + R1 * (1 - a1) * I_k(k);
end

OCV_used = interp1(SOC_breakpoints, OCV_tableData, SOC_CC_k, 'linear', 'extrap');

R0_ideal = 0.0007;
V_meas_ideal = OCV_used(:) - V1_hat(:) - (I_k(:) * R0_ideal);

y_k = OCV_used(:) - V1_hat(:) - V_meas_ideal(:);
x_k = I_k(:);

x_centered = x_k - mean(x_k);
y_centered = y_k - mean(y_k);
[U, S, V] = svd([x_centered, y_centered]);

R0_TLS = -V(1,2) / V(2,2);
R0_datasheet = 0.0007;
SOH_ratio = (R0_datasheet / R0_TLS) * 100;

```

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Bonus 3 Results (TLS Identification):
Estimated R0 (TLS) = 0.000700 Ohms
State of Health (SOH) = 100.00 %
=====

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Results and Conclusion

The resulting behaviors provide key insights into the cell's electro-thermal and dynamic performance:

- **Terminal Voltage Response:** The simulated terminal voltage ($V_{terminal}$) closely tracks the transient load demands. The immediate voltage drops during current spikes clearly reflect the ohmic polarization governed by R_0 , while the subsequent exponential recovery demonstrates the effectiveness of the RC polarization network.
- **SOC Tracking (Coulomb Counting vs. EKF):** The open-loop Coulomb Counting method tracks the capacity accurately under ideal conditions. However, when paired with the EKF observer, the closed-loop filter successfully eliminates tracking drift, demonstrating superior resilience against sensor errors and unmodeled noise disturbances.
- **SOH Identification:** The TLS algorithm successfully extracted the effective internal resistance from the noisy drive-cycle data without bias. The resulting SOH calculation provides an accurate quantitative indicator of the cell's current degradation state, validating the robustness of the identification pipeline for practical Battery Management System (BMS) integration.

